Lesson 146 (2.0) Ground

Lesson Objective

During this lesson, the instructor will evaluate the applicant's aeronautical knowledge on the listed areas of operation based on the current FAA Private Pilot ACS.

Pre-Brief

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Aeronautical Information Manual (AIM).
  - Chapter 01 - Air Navigation.
  - Chapter 02 - Aeronautical Lighting and Other Airport Visual Aids.
  - Chapter 03 - Airspace.
  - Chapter 04 - Air Traffic Control.
  - Chapter 06 - Emergency Procedures.
- Airplane Flying Handbook. (FAA-H-8083-3C)
  - Chapter 11 - Night Operations.
  - Chapter 18 - Emergency Procedures.
- Florida Tech Aviation– VFR Cross-Country Flight Planning Guide
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
  - Chapter 05 - Aerodynamics of Flight.
  - Chapter 07 - Aircraft Systems.
  - Chapter 08 - Flight Instruments.
  - Chapter 10 - Weight and Balance.
  - Chapter 11 - Aircraft Performance.
  - Chapter 12 - Weather Theory.
  - Chapter 13 - Aviation Weather Services.
  - Chapter 14 - Airport Operations.
  - Chapter 15 - Airspace.
  - Chapter 16 - Navigation.
  - Chapter 17 - Aeromedical Factors
- Warrior III Pilot's Information Manual.
  - Section 03 - Emergency Procedures.
  - Section 05 - Performance.
  - Section 06 - Weight and Balance.
  - Section 07 - Systems.

Instructor Guidance

- This lesson is a formal oral knowledge evaluation — not a review session. Your role is examiner, not instructor. Ask questions, listen to the answers, and assess the student's knowledge against ACS standards. Do not teach through the content or fill in gaps as they appear — identify them and address them in the post-brief.
- Use a copy of the Private Pilot ACS as your guide and work through each area of operation systematically. Mark each item as meeting, approaching, or falling short of the standard. This gives you a precise record to share with the student in the post-brief and to reference when determining EOC readiness.
- Pay particular attention to areas that were flagged as deficient in Lesson 144. If the student was assigned specific review homework after that lesson, open with questions in those areas to confirm whether the gaps have been addressed.
- Do not advance the student to the EOC mock in Lesson 149a if meaningful knowledge deficiencies remain after this evaluation. The completion standard requires all knowledge areas to meet the ACS — if they do not, schedule additional ground instruction on specific topics before proceeding.
- During the post-brief, give the student a clear and honest assessment of where they stand. Identify the specific areas that need improvement, assign targeted review, and set a clear expectation for what must be demonstrated before the End-of-Course test.

Lesson Content

Private Pilot Airman Certification Standards

- Ensure that the applicant is familiar with the Private Pilot for Airplane Category Airman Certification Standards (ACS)

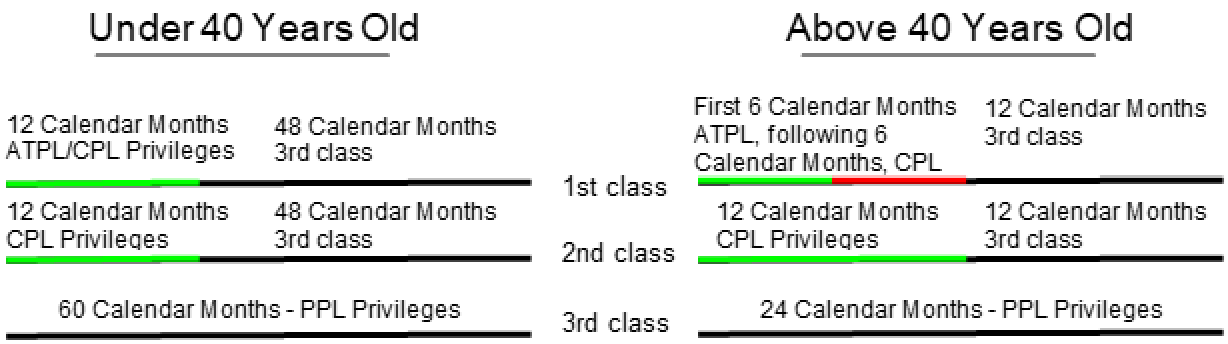
Pilot Qualifications

- Certification Requirements.
    - Eligibility requirements to hold a Private Pilot Certificate.
        - 61.103
            - To be of age 17 or higher. (16 for glider and balloon ratings)
            - To read, speak, write, and understand the English language.
            - To hold a valid US student, sport, or recreational pilot license.
            - To have received an endorsement to take the applicable written exam.
            - To have passed the applicable written exam.
            - To have received an endorsement to take the applicable practical exam.
            - To have the required aeronautical experience that is applicable for the rating sought before submitting the application.
            - To have passed the applicable practical exam.
- Recent Flight Experience.
    - Definitions
        - What is proficiency?

        - Proficiency refers to a pilot's ability to operate an aircraft in a safe and effective manner.
      - What is currency?
        - Currency refers to the legal requirements the FAA sets in order to exercise the privileges of a pilot's certificate.
      - What is the difference?
        - Proficiency does not guarantee currency and currency does not guarantee proficiency.
        - It is the duty of the PIC to be able to determine both when exercising the privileges of their certificate.
  - What does a private pilot need to do in order to be considered current?
      - General Currency
        - The pilot must complete a Flight Review every 24 calendar months.
        - A flight review consists of a 1-hour flight and a 1 hour ground. It is considered complete when the instructor gives the pilot an endorsement certifying their proficiency.
      - Passenger Currency
        - 61.57
        - Within the preceding 90 days the PIC must have completed three takeoff and landings to carry passengers.
        - To carry passengers at night the three landings must be at night and to a full stop.
        - To carry passengers in a tailwheel airplane the landings must have been in a tailwheel airplane and done to a full stop.
- Record Keeping.
  - When do you need to log flights?
      - 61.51
        - Any training activity.
        - Any flight to maintain currency.
        - A flight review.
  - What information do you need to log about each flight?
      - 61.51
        - General Information
          - Date.
          - Flight time.
          - Location. (departure and destination)
          - Type and identification of the Aircraft.
          - The name of the safety pilot if required by 91.109.
        - Type of Piloting Experience
          - Solo.

        - PIC.
        - SIC.
        - Flight and ground training received from an authorized instructor.
      - Condition of Flight
        - Day or night.
        - Actual or simulated instrument.
        - Night vision goggles.
- Privileges and Limitations.
  - What are the privileges associated with a Private Pilot Certificate?
    - 61.113
    - A private pilot may:
      - Act as PIC of an aircraft as per 61.113.
      - Act as PIC of an aircraft for compensation or hire in connection with any business or employment if:
        - (1) The flight is only incidental to that business or employment; and
        - (2) The aircraft does not carry passengers or property for compensation or hire.
      - Act as PIC of an aircraft for a charitable, nonprofit, or community event flight described in 91.146.
      - Be reimbursed for aircraft operating expenses that are directly related to search and location operations sanctioned under
        - A local, State, or Federal agency; or
        - An organization that conducts search and location operations.
      - Demonstrate aircraft in flight to a prospective buyer if the pilot has a minimum of 200 hours of logged flight time.
      - Act as PIC of an aircraft towing a glider provided the requirements in 61.69 are met.
  - What are the limitations associated with a Private Pilot Certificate?
    - 61.113
    - A private pilot may not:
      - Act as PIC for compensation or hire.
      - Pay less than the pro rata share of operating expenses of a flight with passengers.
    - What is a “pro rata” share of operating expenses?
      - This is a proportional split of the operating costs to include fuel, oil, airport expenditures, and rental fees.
- Medical Certificates.
  - Part 67
  - Class.
    - What are the different classes of medical certificate?

        - 1st, 2nd, and 3rd class.
      - What are the privileges associated with each class of medical?
        - 1st class
          - ATPL, CPL, PPL, CFI
        - 2nd class
          - CPL, PPL, CFI
        - 3rd class
          - PPL, CFI
  - Expiration.
    - What are the durations of the medical certificates?



  - Denial of a medical certificate.
    - Eamples of disqualifying conditions.
      - Reference the FAA website.
    - If you are denied a medical based on a disqualifying condition, what course of action may you take?
      - You may appeal the decision within 30 days.
    - When would a Statement of Demonstrated Ability be issued?
      - For a condition that does not have the potential to get worse.
    - Does a SODA need to be renewed?
      - No.
    - When would a Special Issuance be issued?
      - For a condition that does have the potential to get worse.
    - Does a SI need to be renewed?
      - Yes, a SI has a defined end date which can limit the duration of an associated medical.
    - Temporary disqualifications.
      - Who is responsible for determining if a medical is still valid after it is issued, and a new medical condition arises?
    - The pilot who holds the medical.
- Part 68 - Basic Med.
  - AC 68-1

- What do you need to fly under Basic Med?
  - 61.23(c)(3)
  - A valid US driver's license.
  - To have previously held a medical issued under part 67.
  - To have received a comprehensive medical examination from a State-licensed physician during the previous 48 months.
- What are the privileges of flying under Basic Med?
  - 61.113(i)
  - Private pilot privileges.
- What are the limitations of flying under Basic Med?
  - 61.113(i)
  - Private pilot limitations.
  - Additionally:
    - The aircraft must:
      - Be authorized to carry no more than 7 occupants.
      - Have a maximum takeoff weight of no more than 12,500 pounds.
      - Have no more than 6 passengers on board.
    - The flight may not:
      - Exceed 18,000 feet MSL.
      - Exceed 250 knots.
      - Exit the US unless authorized by the country entering.
    - The pilot must have available in their logbook:
      - The completed Comprehensive Medical Examination Checklist. (CMEC)
      - The certificate of course completion required under 61.23(c)(3).
- Who can fly under Basic Med?
  - Pilots exercising Private pilot privileges as Pilot in Command (PIC) or as a required flight crew member (such as a safety pilot).
  - Flight Instructors
  - Pilot Examiners

- Documents are Required to Exercise Private Pilot Privileges.
  - Pilot Certificate.
  - Medical Certificate.
  - Government Issued Photo Identification.
  - Endorsements.
    - High Performance
    - High Altitude
    - Complex
    - Tailwheel

Airworthiness Requirements

- General Airworthiness Requirements and Compliance.
  - Certificate location and expiration dates.
  - Airworthiness information.
    - Define airworthiness.
      - The aircraft must meet the type design.
      - Have all required inspections.
      - Comply with all airworthiness directives.
      - Be in a safe condition to fly.
    - Who is responsible for determining the airworthiness of an aircraft to be flown?
      - The Pilot in Command.
    - Who is responsible for maintaining an aircraft in an airworthy condition?
      - The owner or operator of the aircraft.
  - Type Design.
    - What is a type design?
      - The type design is a document that outlines all of the specifications relevant to manufacturing the aircraft.
    - Can you access the type design?
      - No, the information is proprietary to the manufacturer.
    - What do you have access to as a pilot to ensure the aircraft meets its type design?
      - The type certificate data sheet.
      - This is a document containing all information in the type design that is relevant to the pilot.
    - Ensure the student knows where to find the TCDS.
      - The FAA website.
  - Required inspections and airplane logbook documentation.
    - What are the required Inspections for the aircraft to be airworthy?
      - **AVIATES**
      - **A**nnual.
        - 91.409
        - Every 12 calendar months.
        - The annual inspection must be conducted by an Airframe and Powerplant (A&P) with Inspector Authorization (IA).
      - **V**OR.
        - 91.171
        - Within the preceding 30 days before the flight.
        - IFR requirement.
      - **1**00 hour Inspection.
        - 91.409
        - Required every 100 hours of time spent in the air.

            - Only required with operations for-hire.
            - Can overfly by 10 hours to get to maintenance.
                - Overflying takes away time from the next 100 hour inspection.
                - Only for the purpose of transporting it to maintenance.
        - **A**ltimeter.
            - 91.411
            - Every 24 calendar months.
            - IFR requirement.
        - **T**ransponder.
            - 91.413
            - Every 24 calendar months.
            - Only for operations that require a transponder.
            - If you can't get this inspection, can you fly the aircraft at all?
                - Yes, you must keep the transponder off and remain only where a transponder is not required.
        - **E**LT.
            - 91.207
            - Every 12 calendar months.
            - The batteries must be either recharged or replaced after
                - One hour of cumulative use
                - Or 50 percent of the batteries useful life has expired
        - **S**tatic system.
            - 91.411
            - Every 24 calendar months.
            - IFR requirement.
    - What are the different sections of the maintenance log?
        - Airframe
        - Powerplant
        - Propeller
        - Avionics
        - Airworthiness Directives
- Airworthiness Directives (ADs) and Special Airworthiness Information Bulletins. (SAIBs)
    - Airworthiness directives:
        - Define what an airworthiness directive is.
            - A regulatory notice from the FAA about the airworthiness of an aircraft.
        - What are the types of ADs?
            - Notice of proposed rulemaking. (NPRM)
            - Final rule.
            - Emergency.

                - What are the methods of compliance with ADs?
                    - One-time.
                    - Recurring.
            - Special Airworthiness Information Bulletins. (SAIBs)
                - What is a Special Airworthiness Information Bulletin?
                    - A non-regulatory notice from the FAA about the airworthiness of an aircraft.
                    - They will be issued when a defect does not warrant an AD but the owners of an aircraft should know about an issue.
            - Service Bulletins (SBs).
                - What is a service bulletin?
                    - A non-regulatory notice from the manufacturer about the airworthiness of an aircraft.
    - Purpose and procedure for obtaining a special flight permit.
        - What is a special flight permit and when would one be needed?
        - What are the limitations of a special flight permit?
        - If you must divert from the route cleared in the special flight permit, can you continue the flight after landing at the airport you diverted to?
        - How would a pilot obtain a special flight permit?
- Pilot Performed Preventive Maintenance.
    - Is a private pilot allowed to perform any type of maintenance on an aircraft?
        - [43.3(g)]
    - What preventative maintenance is a private pilot allowed to do?
        - [Part 43(c) Appendix A]
    - What records must be kept after this maintenance?
        - [43.9]
- Requirements for Day and Night VFR Flight.
    - What are the documents that must be in the aircraft for it to be considered airworthy?
        - **A**irworthiness certificate.
            - [91.203]
            - [The FAA website.]
            - It does not expire but it may be rendered invalid if the aircraft is not airworthy.
            - It is important to check the tail number and signature during a pre-flight inspection.
            - The airworthiness certificate must be left visible. Do not allow the registration certificate to cover it.
        - **R**egistration certificate.
            - [91.203]
            - [Part 47]

- Typically expires every 7 years.
  - Where can you find information on the registration process?
    - On the FAA website
- Check tail number and expiration date during pre-flight.
- Different ways to invalidate the registration certificate:
  - Death of owner.
  - Transfer of ownership.
  - The aircraft is found doing illegal activities.
  - The aircraft is totaled.
  - The certificate holder requests in writing to cancel their registration.
  - The certificate holder loses US Citizenship.

- **R**adio station license / Radio operator permit.
  - ICAO Article 29
  - This is not an FAA requirement, it is required by foreign governments.
  - Obtained through the Federal Communications Commission (FCC).
  - Radio station license is for the airplane.
    - Expires every 10 years.
  - Radio operator permit is for the pilot.
    - Does not expire.
- **O**perating limitations.
  - In FAA Approved Airplane Flight Manual (AFM) section 02.
  - Required by a combination of the following regulations:
    - 21.5.
    - 91.9.
    - 23.2620.
- **W**eight and balance.
  - FAA Approved AFM (POH) section 06.
  - If required by:
    - 21.5.
    - 23.2620.
    - The TCDS.
- **E**quipment list
  - This refers to an updated list of all the equipment in the aircraft including the weight, arm, and usually the serial number, and part number of each component.
  - If required by:
    - The TCDS.
    - Supplemental Type Certificate.
- **S**upplements
  - **F**AA form 337.

            - Major alterations or repairs.
            - Examples of major alterations can be found in Part 43.
        - **L**ife-limited parts list.
            - A list of any part for which a mandatory replacement limit is specified in the type design or the maintenance manual.
        - **I**noperative equipment list.
            - Refer to "Flying with inoperative equipment" found below.
        - **C**ompass deviation card.
        - **E**xternal data plate.
            - Located on the left side of the fuselage, by the tail.
        - **S**upplemental type certificate.
- What is the required equipment the aircraft must have?
    - Day VFR required equipment (ATOMATOFLAMES)
        - 91.205
        - **A**irspeed Indicator.
        - **T**achometer.
            - For each engine.
        - **O**il Pressure Gauge.
            - For each engine using pressure system.
        - **M**anifold Pressure Gauge.
            - For each altitude engine.
            - Define and give an example of an altitude engine.
        - **A**ltimeter.
        - **T**emperature Gauge.
            - For each liquid cooled engine.
        - **O**il Temperature Gauge.
            - For every air-cooled engine.
        - **F**uel Quantity Gauge.
            - For each fuel tank.
        - **L**anding Gear Position Lights.
            - If retractable.
        - **A**nti-Collision Lights.
            - If the aircraft is certificated after March 11th, 1996.
        - **M**agnetic Direction Indicator.
        - **E**LT.
            - 91.207
        - **S**eatbelts and shoulder harnesses.
    - Night VFR required instruments (FLAPS).
        - 91.205
        - **F**uses/resettable circuit breakers.

        - **L**anding Light (Electric)
            - for hire operations.
        - **A**nti-Collision Lights.
            - Aircraft certified after August 11th, 1971.
            - Aircraft certified before that date STILL NEED anti-collision lights but the standards/requirements are different.
        - **P**osition Lights.
        - **S**ource of Electricity (Alternator/Battery).
- Flying with inoperative equipment.
    - How do you determine airworthiness when there is inoperative equipment on the aircraft?
        - 91.213
            - If an aircraft has a Minimum Equipment List, then refer to that first and find out if the aircraft is still airworthy.
                - An MEL will list items that may be inoperative without rendering the aircraft unairworthy.
                - The listed items will generally come with limitations on the flight if said item is inoperative.
            - Determining Airworthiness with inoperative equipment in absence of an MEL (TARKY):
                - **T**ype certificate data sheet (TCDS).
                - **A**irworthiness Directives.
                - **R**equired by regulations (part 91).
                - **K**OEL, Kinds of Operating Equipment List (section 2 of POH).
                - **Y**ou
                - If the piece of equipment is NOT required by any of the above:
                    - Remove it from the airplane in accordance with 43.9 or deactivate and placard inoperative.
    - Kinds of Operation Equipment List. (KOEL)
        - A KOEL is a list found in the aircraft's FAA Approved Airplane Flight Manual (POH) Section 2.
        - This list contains all of the equipment that is required for the aircraft to be airworthy, and it is set by the manufacturer and approved by the FAA.
    - Required discrepancy records or placards.
        - If an inoperative piece of equipment is determined to not compromise airworthiness, then it shall be removed from the airplane in accordance to 43.9 or deactivate and placard inoperative.

Weather Information

- Sources of Weather Information for Flight Planning Purposes.
    - Sources that fulfill the legal preflight requirements of 91.103.
        - Calling a weather briefer.
        - 1800wxbrief.com.
        - ForeFlight generated weather briefing.
        - Aviation Weather Center.
- Acceptable Weather Products and Resources.
    - Aviation Routine Weather Report (METAR)
        - AIM 7-1-28 and 7-1-29 details the shorthand used in various ICAO standardized weather reports.
        - Live METARs
        - Info page on AWC
        - What is a METAR?
            - An aviation routine weather report.
            - It is an observation of weather that is happening at the time of the report.
        - How long is a METAR valid for?
            - 1 hour.
        - How often is a METAR issued?
            - Every hour, 55 minutes past the hour.
        - Aviation Selected Special Weather Report (SPECI)
            - Under what circumstances a SPECI will be issued?
    - Terminal Aerodrome Forecast (TAF)
        - Live TAFs
        - Info page on AWC.
        - What is a TAF?
            - A terminal aerodrome forecast.
            - It is a forecast of expected conditions to occur for a time period after the issuance.
        - How long is a TAF valid for?
            - 24 or 30 hours, as depicted on the product.
        - How often is a TAF issued?
            - Every 6 hours.
    - Winds/Temps Data
        - AIM 8-3-1 details FB Winds
        - Live FB Winds
        - Info page on AWC.
        - What are FB Winds?
            - FB winds are a forecast of the winds and temperatures aloft.
        - How often are FB Winds issued?

- Every 6 hours at 0000Z, 0600Z, 1200Z, and 1800Z
    - How far out does each issuance forecast?
        - Each issuance has a forecast for 6, 12, and 24 hours.
    - How long is each issuance valid for?
        - The valid times are printed on the chart.
  - PIREP
    - AIM 7-1-18,20,21,22,23
    - Live PIREPs
    - What is a PIREP?
        - A pilot report of weather in flight.
    - How often are they issued?
        - Whenever a pilot deems necessary.
        - Urgent (UAA) vs Routine (UA).
  - Prognostic charts
    - Low Level Significant Weather Chart (Surface to FL240)
        - Info page on AWC.
    - Surface Prognostics Chart
        - Info page on AWC.
  - Adverse weather
    - WA.
    - WS.
    - WST.
  - Freezing level chart
    - Info page on AWC.
  - Graphical Forecast for Aviation (GFA) tool
    - Info page on AWC.
- Meteorology Applicable Under Visual Meteorological Conditions. (VMC)
  - Atmospheric composition and stability.
    - What is atmospheric stability?
        - The ability for air in the atmosphere to return to its original state when displaced.
    - What is the general composition of the atmosphere?
        - 78% Nitrogen
        - 21% Oxygen
        - 1% Other
    - Does the composition of the atmosphere change with altitude?
        - While temperature and pressure change with altitude, the composition of the atmosphere does not.
    - What is an air mass?
        - A section of air that has similar properties.

        - What is a Front?
            - The boundary between two or more air masses.
- Weather associated with various fronts.
    - Warm Fronts
        - Occurs when the warm mass of air advances and replaces a body of colder air. Warm fronts move slowly, 10-25 mph. Slope of the advancing front slides over the top of the cooler air and gradually pushes it out of the area. As the warm air is lifted, the temperature drops, and condensation occurs from the humidity of the air.
        - Generally, cirriform, stratiform clouds (stable air) along with fog can be expected to form along the frontal boundary. (rain evaporates when it gets close to the cooler ground).
        - Light to moderate steady precipitation.
        - Flight toward a warm front:
            - Visibility decreases.
            - Falling barometric pressure.
            - Weather overall deteriorates.
            - May be impossible to continue a VFR flight after flying through a warm front.
    - Cold Fronts
        - Occurs when a mass of cold, dense and stable air advances and replaces a body of warmer air.
        - Moves at 25-30 mph, can move as fast as 60 mph.
        - Rapidly ascending air causes the temperature to decrease suddenly, forcing creation of clouds.
        - It lifts moist air and creates instability.
            - This contributes to two of the three criteria for a thunderstorm.
        - Prior to cold front passing, cirriform towering cumulus clouds are present, and cumulonimbus clouds may develop.
        - Showery precipitation.
        - After frontal passage, clouds dissipate to cumulus clouds with a corresponding decrease in precipitation.
        - Good visibility prevails.
        - Flight toward a cold front:
            - Flight progresses to the oncoming cold front; the clouds show signs of vertical development with a broken layer.
            - Pilots should use sound judgment on frontal passage.
    - Stationary Fronts

            - When the forces of two air masses are relatively equal, the boundary layer or front that separates them remains stationary and influences the local weather for days.
        - Occluded Fronts
            - Fast moving cold front catches up with a slow-moving warm front. Warm front weather prevails but is followed by cold front weather.
        - What are frontogenesis and frontolysis?
            - Frontogenesis: the initial formation of a front.
            - Frontolysis: the dissipation of a front.
    - Wind.
        - How does air flow relative to the isobars at the surface? And aloft?
            - Air flows at an angle to the isobars at the surface.
            - Air flows parallel to the isobars aloft.
        - What does a tight pressure gradient indicate in terms of wind?
            - The closer together the isobars the stronger the winds should be in theory.
        - What is wind shear?
            - It is a dramatic shift in wind speed and/or direction.
            - It can exist in a horizontal or vertical direction and occasionally in both.
        - What qualifies as low-level wind shear?
            - Wind shear occurring at or below 2000 feet AGL.
    - Temperature.
        - What is the temperature lapse rate on a standard day?
            - -2 degrees Celsius for every 1000-foot increase in altitude.
        - What is a temperature inversion?
            - When temperature increases with altitude.
        - When might a temperature inversion be experienced?
            - On calm cool nights due to terrestrial cooling of the air at the surface.
            - At some frontal boundaries.
    - Moisture/precipitation.
        - What are the different intensities of precipitation?
            - Light, Moderate, Heavy, and Extreme.
    - Weather system formation, including air masses and fronts.
        - What is the source of all weather?
            - The source of all weather is the uneven heating of the earth's surface by the sun.
    - Clouds.
        - Define ceiling.
            - The lowest layer of broken or overcast clouds, or the vertical visibility.
        - Clouds are classified by their height, shape, and characteristics.
            - Ensure the applicant is familiar with and can identify various different types of

clouds.

- Turbulence.
    - Refer to [TBL 7-1-10](#) in [AIM 7-1-21](#)
    - What is the difference between turbulence and chop?
        - Turbulence
            - Turbulence is a random, sporadic movement of the aircraft.
        - Chop
            - Chop is a rapid and somewhat rhythmic movement of the aircraft.
    - What are the intensities of turbulence or chop?
        - Light
        - Moderate
        - Severe
        - Extreme
    - What are the frequencies of turbulence?
        - Occasional
        - Intermittent
        - Continuous
- Thunderstorms and Microbursts.
    - What are the criteria required for the formation of a thunderstorm?
        - Lifting action.
        - Moisture.
        - Unstable air.
    - What are the stages of thunderstorm?
        - Developing / Cumulus
            - Towering cumulus (rising air), little rain, occasional lightning.
        - Mature
            - fully formed, heavy rain, frequent lightning, strong winds.
            - Starts when rain makes contact with the surface.
        - Dissipating
            - downdrafts, rainfall decreases, still producing strong winds and lightning.
    - Different types of thunderstorms:
        - Air-mass
            - Generally, a result of surface heating.
            - Typically, last up to two hours or less.
            - Generally, of moderate severity, yet still incredibly hazardous.
        - Steady State
            - Generally, a result of frontal activity, troughs, or converging winds resulting in a lifting action.
            - These have more potential to be severe thunderstorms.
        - Squall line

            - A narrow band of TS, usually ahead of a cold front (moist, unstable air, lifting action). May also be completely independent of a front.
            - Typically, quite severe and wide.
        - Embedded
            - An embedded thunderstorm is simply a thunderstorm obscured from view by the presence of clouds or other weather phenomena.
        - Single cell
            - develop on warm humid days. severe, produce hail, and microburst.
        - Multi-cell
            - develop in clusters, cover large areas, and can move in different directions.
        - Supercell
            - Single long-lived TS. large hail and significant tornadoes.
    - What weather to expect in a thunderstorm:
        - Lightning.
        - Turbulence.
        - Hail.
        - Tornadoes.
        - Wind shear.
        - Icing.
    - Microburst
        - Occurs in a space of less than one mile horizontally and within 1000 feet vertically.
        - Can span for 15 min.
        - Produce downdrafts of 6,000 feet per minute.
        - Can produce hazardous wind direction change of 45 degrees or more in a matter of seconds.
        - The plane will encounter an increase in headwind, then performance decreasing downdrafts, then a shear tailwind, and can result in terrain impact.
        - It happens in confined areas.
    - Where are microbursts most likely to occur?
- Icing and freezing level information.
    - AIM 7-1-19 and 7-1-20
    - What are the performance impacts of icing?
        - Drag increases.
        - lift decreases.
        - Weight increases.
        - Thrust decreases.
        - Stall speed increases.
    - What are the different categories of icing in general?

        - Structural
            - Requires the presence of visible moisture and for the aircrafts surface to be at or below freezing temperature.
        - Induction
            - Icing of the induction system (carburetor)
                - Can happen with temperatures as high as 70F and a relative humidity of 80%.
                - Fuel vaporizes in the venturi, causing the temperature of the air and carburetor walls to decrease, which allows for the possible formation of ice.
            - What can the pilot do when dealing with induction icing?
                - Turn carb heat on
                    - Initially there will be a small reduction in RPM as warm air is less dense than cold air and this enriches the mixture.
                    - As time passes RPM should steadily increase as ice inside of the carburetor melts.
        - Instrument
            - Pitot tube and static ports.
            - Antennas.
            - Stall warning tab.
            - OAT probe.
    - What are the different kinds of structural icing?
        - Clear
        - Rime
        - Mixed
    - What are the different levels of icing accumulation rates?
        - Trace.
        - Light.
        - Moderate.
        - Severe.
- Fog/mist.
    - Occurs when the temperature of the air near the ground is cooled to the it's dew point.
    - WMO Fog
    - Weather.gov Fog
    - Types (reported when visibility is less than 5/8th of a mile):
        - Radiation Fog
            - Occurs when the ground cools rapidly due to terrestrial radiation, and surrounding air reaches dew point. As the sun rises and temperature increases, the fog burns off.
        - Advection Fog

        - when a layer of warm moist air moves over a cold surface. A 15-knot wind is necessary. Common to coastal areas where sea breezes can blow the air over cooler landmasses.
      - Upslope fog
        - Occurs when moist, stable air is forced up sloping land features like a mountain range. Also requires wind for formation and continued existence.
      - Steam fog
        - Forms when cold dry air moves over warm water. As water evaporates it turns into steam.
      - Precipitation fog
        - Occurs when the raindrops are warmer than the surrounding air.
  - Frost.
    - Ice crystal deposits are formed by deposition (gas to solid) when temp and dew point are below freezing.
- Flight Deck Displays of Digital Weather and Aeronautical Information.
  - Not applicable to our systems.

Cross-Country Flight Planning

- Route and Altitude Planning.
  - Airspace classes.
  - Special Use Airspace (SUA).
  - Navigation/communication systems and facilities.
  - Terrain and Obstacles.
  - VFR cruising altitudes.
  - Wind.
- Calculating:
  - Time, climb and descent rates, course, distance, heading, true airspeed, and groundspeed.
  - Estimated time of arrival to include conversion to universal coordinated time (UTC).
  - Fuel requirements, to include reserve.
- Elements of a VFR Flight Plan.
- Procedures for Activating and Closing a VFR Flight Plan.
    - Pilotage and Dead Reckoning
  - Magnetic Compass Errors.
  - Topography.
  - Selection of Appropriate:
    - Route.
    - Altitude.
    - Airplane.
    - Plotting a Course, to Include:

        - Determining heading, speed, and course.
        - Wind correction angle.
        - Estimating time, speed, and distance.
        - True airspeed and density altitude.
    - Power Setting Selection.
    - Planned Versus Actual Flight Plan Calculations and Required Corrections.
- Navigation Systems and Radar Services
    - Ground-Based Navigation
    - Satellite-Based Navigation
    - Radar Assistance to VFR Aircraft
    - Transponders
- Lost Procedures
    - Methods to Determine Position
    - Assistance Available if lost

National Airspace System

- Types of Airspace and Airspace Classes.
    - Controlled Airspace - Airspace within which Air Traffic Control (ATC) service is provided.
        - AIM 3-2
        - Class A
            - 91.135
            - "**A**bove"
            - Airspace from 18,000 feet Mean Sea Level (MSL) up to and including 60,000 feet MSL.
            - Operations conducted under Instrument Flight Rules. (IFR)
            - It extends 12 miles offshore of the 48 contiguous states and Alaska.
        - Class B
            - 91.131
            - "**B**usiest"
            - Generally, airspace from the surface up to 10,000 feet MSL around the busiest airports in the nation. (total operations or passengers carried)
            - The configuration will be slightly different for every airport.
            - The airspace may have multiple layers, resembling an upside-down wedding cake.
            - ATC provides separation services within the airspace.
            - Some airports do not allow student, sport, or recreational pilots to land, usually due to the high traffic volume.
                - Section 4 of appendix D of Part 91.
        - Class C
            - 91.130

        - "**C**ongested"
        - Designated for airports less busy than Bravo airports, but still with a significant volume of traffic.
        - Generally, the first layer extends from the surface up to 4,000 feet above the airport's surface with a 5NM radius. (depicted in MSL)
        - A second layer usually extends from 1,200 feet to 4,000 feet above airport elevation with a 10NM radius.
        - The exact dimensions of the airspace may be different for each airport, and some airports may also have another outer area.
    - Class D
        - 91.129
        - "**D**reary"
        - Generally, airspace from the surface up to 2,500 feet above airport elevation. (depicted in MSL)
        - It surrounds an airport with a control tower. The control tower may be operational full time or part time.
        - In the case of part time operations, the airspace reverts to either Class E or Class G airspace when the tower is closed. More information about this is located in the chart supplement page for the specific airport.
        - May be individually tailored for each airport.
        - Usually designed to contain instrument approaches at the airport.
    - Class E
        - 91.127
        - "**E**verywhere else"
        - Controlled airspace that is not class A, B, C, or D.
        - It extends up from either the surface or a designated altitude to the overlying or adjacent controlled airspace.
        - Unless otherwise designated, it begins at 14,500 feet MSL over the 48 contiguous states and Alaska.
        - Usually, it begins at either 700 or 1,200 feet above the surface in areas not surrounding an airport for transition to and operations in the en route environment.
        - It goes up to but does not include 18,000 feet MSL.
        - Generally, it extends 12 NM offshore.
        - The airspace above 60,000 feet MSL.
- Uncontrolled Airspace - Airspace within which Air Traffic Control (ATC) services are not provided.
    - Class G
        - 91.126
        - AIM 3-3

            - "**G**round"
            - The portion of the airspace system that is not designated as A, B, C, D, or E.
            - Generally, from the surface up to the overlying controlled airspace.
            - Visual Flight Rules (VFR) apply.
            - ATC has no authority or responsibility.
- Associated Requirements and Limitations when Entering and/or Operating Inside of the Different Classes of Airspace.
    - AIM 3-2
    - Class A.
        - Established two-way radio communication.
        - Aircraft must be IFR equipped.
        - ADS-B out.
        - ATC clearance.
        - No visibility or cloud clearance requirements.
    - Class B.
        - Established two-way radio communication.
        - Transponder with altimeter encoding. (Mode C or better)
        - ADS-B out.
        - ATC clearance.
        - Visibility of 3SM.
        - The aircraft must remain clear of clouds.
    - Class C.
        - Established two-way radio communication.
        - Transponder with altimeter encoding. (Mode C or better)
        - ADS-B out.
        - Visibility of 3SM.
        - The aircraft must remain 1,000' above, 500' below and 2,000' horizontally from all clouds.
    - Class D.
        - Established two-way radio communication.
        - Visibility of 3SM.
        - The aircraft must remain 1,000' above, 500' below and 2,000' horizontally from all clouds.
    - Class E.
        - Below 10,000 feet MSL
            - Visibility of 3SM.
            - The aircraft must remain 1,000' above, 500' below and 2,000' horizontally from all clouds.
        - At or above 10,000 feet MSL
            - Visibility of 5SM.

            - The aircraft must remain 1,000' above, 1,000' below and 1SM horizontally from all clouds.
    - Class G.
        - [91.126]
        - [AIM 3-3]
        - At or below 1,200 feet Above Ground Level (AGL)
            - Day
                - Visibility of 1SM.
                - The aircraft must remain clear of clouds.
            - Night
                - Visibility of 3SM.
                - The aircraft must remain 1,000' above, 500' below and 2,000' horizontally from all clouds.
        - Above 1,200 feet AGL but less than 10,000 feet MSL
            - Day
                - Visibility of 1SM.
                - The aircraft must remain 1000' above, 500' below and 2000' horizontally from all clouds.
            - Night
                - Visibility of 3SM.
                - The aircraft must remain 1000' above, 500' below and 2000' horizontally from all clouds.
        - Above 1,200 feet AGL and at or above 10,000 feet MSL.
            - Visibility of 5SM.
            - The aircraft must remain 1,000' above, 1,000' below and 1SM horizontally from all clouds.
    - Additional Requirements or Limitations.
        - Transponder / ADS-B Out requirement: 91.215.
            - Class A, B, C.
            - Mode C veil (Surrounding Class B airspace).
            - Below class B or class C.
            - At or above 10, 000 ft.
        - ATC authorized deviations.
- Charting Symbology.
    - Go through a sectional with the applicant and use Situation Based Training (SBT) to determine that their knowledge meets or exceeds the ACS requirements of a private pilot.
    - An emphasis should be placed on the ability to figure out an answer when the applicant does not know the answer.
        - How to use the legend on the sectional and the Aeronautical Chart Users' Guide.
- Special Use Airspace. (SUA)

- AIM 3-4
- **WARMPC**
  - **W**arning Areas.
    - It is similar to a restricted area.
    - The government does not have jurisdiction over the airspace.
    - The airspace has defined dimensions, extending from 3NM offshore of the U.S.
    - It contains activity that may be hazardous to nonparticipating aircraft.
    - It is used to caution non-participating pilots.
    - It is depicted with a "W" followed by a number on the sectional chart.
  - **A**lert Areas.
    - Areas where there is a high volume of pilot training or an unusual type of aerial activity.
    - Pilots are responsible for collision avoidance.
    - It is depicted with an "A" followed by a number on the sectional chart.
  - **R**estricted Areas.
    - Areas in which operations are hazardous to non-participating aircraft.
    - They contain artillery firing, guided missiles, etc.
    - The flight of aircraft may be restricted, but not prohibited.
    - To enter an active restricted area, the pilot must have prior permission from the controlling agency.
    - It is depicted with an "R" followed by a number on the sectional chart.
    - Additional information may be found on the back of the sectional chart, including active times.
  - **M**ilitary Operations Areas. (MOAs)
    - Separates certain military training activities from IFR traffic.
    - No live artillery is fired in MOAs.
    - IFR traffic may be cleared through the airspace if ATC can provide separation.
    - Depicted on aeronautical charts with defined vertical and lateral limits.
    - It is depicted with a name on the sectional chart.
    - Additional information for each may be found on the back of the sectional chart.
  - **P**rohibited Areas.
    - Airspace in which flight is prohibited.
    - The purpose of them is usually for national security or welfare.
    - It is depicted as a "P" followed by a number on the sectional chart.
    - Camp David, White House, Congressional buildings.
  - **C**ontrolled Firing Areas. (CFAs)
    - They contain activity that is hazardous to non-participating aircraft.
    - The activity will stop when the spotter sees non-participating aircraft.

- They are published in the chart supplement under Special Notices.

- Other Airspace Areas.
  - AIM 3-5
  - Temporary Flight Restrictions. (TFRs)
    - Designated by the Flight Data Center Notice to Air Missions (FDC NOTAMs)
      - Begins with phrase "FLIGHT RESTRICTIONS" followed by location, effective time period, and defined in statute miles, and altitudes affected
    - Purposes may include:
      - Protect persons or property in air or on surface from existing or imminent hazard
      - Provide safe environment for operation of disaster relief aircraft
      - Prevent unsafe congestion of sightseeing aircraft above incident or event
      - Protect declared national disasters for humanitarian reasons in Hawaii
      - Protect the President, Vice President, or other public figures
      - Provide a safe environment for space agency operations
    - If penetrated, pilot may undergo security investigations and certificate suspensions
  - Parachute Jump Areas (PJAs)
    - Airports at which there are high numbers of parachute operations
    - Published in chart supplement under Parachute Jumping Areas (page 496) and depicted on sectional charts
  - Published VFR Routes
    - Used for transitioning around, under, or through complex airspace
    - Usually found on VFR terminal area planning charts
    - Often referred to as VFR flyway, VFR corridor, Class B airspace VFR transition route, and terminal area VFR route.
  - Terminal Radar Service Areas (TRSAs)
    - Areas in which pilots can receive additional radar services
    - Purpose is to provide separation for IFR and participating VFR traffic
    - The airspace within the TRSA becomes Class D around the airports
    - Remaining portion overlies other controlled airspace, which is normally Class E
    - Depicted on VFR sectional charts and terminal area charts with a solid black line and altitudes for each segment
    - Class D portion charted with a blue segmented line
    - Participation is voluntary and encouraged
  - Special Air Traffic Rules (SATR) and Special Flight Rules Area. (SFRA)
    - An area designated by the FAA to have additional rules and regulations which must be followed while operating inside of.
    - Some examples of SATRs are in Destin Florida
    - Some examples of SFRAs are in Washington DC, and The Grand Canyon.
    - There may be additional training required, or you may be required to communicate

with ATC in situations where normally you would not need to.
        - It is critically important that a pilot makes themself aware of all rules that might apply inside of a SFRA or SATR before making a flight inside of one.
    - Military Training Routes (MTRs)
        - Used by military aircraft to maintain proficiency in tactical flying
        - Usually established below 10,000 feet MSL for operations exceeding 250 knots
        - Routes depicted as IR (if IFR) and VR (if VFR), followed by a number
            - Those not exceeding 1,500 feet AGL contain 4 numbers (IR 1206)
            - Those above 1,500 feet AGL contain 3 numbers (VR 207)
    - Wildlife Refuge Area
        - From the surface to 2,000 feet
- Additional Regulatory Concerns.
    - Right of Way Rules (91.113)
        - Who has the most priority over anyone else?
            - In distress aircraft have the right of way over all other aircraft.
        - Least maneuverable category has right of way:
            - **BGAAR**
                - **B**alloon.
                - **G**lider.
                - **A**irship.
                - **A**irplane.
                - **R**otorcraft.
        - What should pilots do if two aircraft are head on?
            - Both divert course to the right.
        - What about converging?
            - Aircraft on the right has the right of way and the aircraft on the left should maneuver to avoid collision
            - See and AVOID
        - Overtaking
            - Aircraft being overtaken has the right of way. When overtaking an aircraft, alter course to the right.
        - Landing:
            - The aircraft at the lower altitude has the right of way. (This rule to cannot be used to purposely cut in front of another plane)
    - Operating Near Other Aircraft (91.111)
        - (a) No person may operate an aircraft so close to another aircraft as to create a collision hazard.
        - (b) No person may operate an aircraft in formation flight except by arrangement with the pilot in command of each aircraft in the formation.
        - (c) No person may operate an aircraft, carrying passengers for hire, in formation

flight.

Performance and Limitations

- The Use of Charts, Tables, and Data to Determine Performance.
    - What are the V speeds of our aircraft?
        - Vx
            - The best angle of climb.
            - Will result in the most altitude gained for any given distance.
            - 63 kts.
        - Vy
            - The best rate of climb.
            - Will result in the most altitude gained for and given time.
            - 79 kts.
        - Vs0
            - Stall speed with a load factor of 1 with flaps set to 40.
            - 44 kts.
        - Vs1
            - Stall speed with a load factor of 1 with flaps set to 0.
            - 50 kts.
        - Vfe
            - Maximum flap extension speed.
            - Extending the flaps above this speed could damage them.
            - 103 kts.
        - Vno
            - Maximum structural cruising speed or normal operating range.
            - This is the maximum safe speed the aircraft should be flown unless in smooth air.
            - 126
        - Vne
            - Never exceed speed.
            - Flight above this speed will likely result in damage to the aircraft.
            - 160
    - What is Va?
    - Does Va change with weight? Why?
- Factors Affecting Performance.
    - Atmospheric conditions.
        - What is pressure altitude?
        - How can you figure out the pressure altitude?
        - What is density altitude?
        - How can you figure out the density altitude?

        - What is the significance of density altitude?
        - On a standard day, would pressure altitude be higher at KMLB or KDEN?
        - On a standard day, would density altitude be higher at KMLB or KDEN?
    - Pilot technique.
        - What impact would improper pilot technique have on the aircraft's performance as read from the charts?
        - Is it possible for a pilot's technique to allow for aircraft performance that is better than listed in the aircraft's performance charts?
    - Airplane configuration.
        - How do flaps impact takeoff ground roll?
    - Airport environment.
        - What are some runway conditions that could impact performance?
            - Runway Slope.
            - Uneven or rough surface.
            - Water or other weather on the surface.
    - Aircraft loading.
        - What is Basic Empty Weight?
        - What is "arm?"
        - What is "moment?"
        - What is "Center of Gravity?" (CG)
        - How do you calculate CG for an aircraft with given weights and arms?
        - How does CG impact performance?
            - Forward
                - Increase Ground Roll
                - Decreased Cruise Speed
                - Stall Speed Increases
                - Easier Stall Recovery
            - Aft
                - Decreased Ground Roll
                - Increased Cruise Speed
                - Stall Speed Decreases
                - Harder Stall Recovery
    - Weight and Balance.
        - How does weight impact performance?
- **Aerodynamics**
    - Refer to Lesson 106

Operations of Systems

- Aircraft structure
    - What is the material of the aircraft skin?

- What kind of structure does the aircraft have?
- What is the aircraft structure composed of?
- Primary flight controls.
    - What are the primary flight controls of our aircraft?
        - Ailerons.
        - Stabilator.
        - Rudder.
    - What axis of motion does each primary flight control manipulate?
        - Ailerons
            - Roll.
            - Motion about the longitudinal axis.
        - Stabilator.
            - Pitch.
            - Motion about the lateral axis.
        - Rudder.
            - Yaw.
            - Motion about the vertical axis.
    - What type of ailerons do we have?
        - Differential ailerons.
        - These are ailerons that deflect more upwards then downwards.
    - Why do we have differential ailerons?
        - To combat adverse yaw.
    - What is adverse yaw?
        - A yawing moment created in the opposite direction of the turn.
        - This is caused by the increased lift on the high wing in a turn, which results in the production of more induced drag on said wing.
        - The increase in upward deflection of the low wing's aileron compensated for this asymmetric drag by increasing parasite drag on the low wing in a turn.
    - What is the difference between a stabilator and an elevator?
        - An elevator is the small portion that moves and is usually attached to a horizontal stabilizer.
        - A stabilator integrates these two components into one large moving control surface.
    - What is an anti-servo tab and what purpose does it have?
        - The anti-servo tab exists to increase the required control pressures at the yoke. This results in less over-control of the aircraft.
- Secondary flight controls
    - What are the secondary flight controls?
        - Flaps
        - Trim
    - What is the purpose of each secondary flight control?

- What are the different kinds of flaps?
- What kind of flaps do we have?
- What are the advantages of our flaps?
- Which of these is the least efficient flap?

- Powerplant and propeller
  - How does the fuel air mixture get created and transported to the engine?
    - Inside of the carburetor and through the induction system
      - Ensure the student can draw this process and understands it fully.
  - How does the exhaust gas leave the engine?
    - Through the exhaust system
      - Ensure the student can draw this process and understands it fully.
  - Tell me what you know about the powerplant of our aircraft.
    - Students should cover the CL4HAND acronym.
    - Students should know what O-320-D3G means.
- Landing gear
  - What type of landing gear system does the Warrior have?
  - What type of struts does the Warrior have?
  - How much should the main struts be extended? What about the nose strut?
  - How inflated should the nose and main tires be?
  - What do you check during the pre-flight?
- Fuel
  - The student should draw the fuel system on the board.
  - How many gallons of fuel can the Warrior hold? Usable?
  - What's the purpose of the gascolator?
  - Why does the system have two fuel pumps?
  - Why does the primer only prime 3 cylinders?
  - How does the carburetor work?
- Oil
  - What is the operating range for the oil quantity?
  - What are the types of oil and when they are used?
  - What is the difference between wet-type and dry-type oil sumps? What does our aircraft have?
  - What are the functions of the oil?
- Hydraulic.
  - What does the hydraulic system do?
    - The hydraulic system is used to actuate the brakes.
  - How many cylinders does the system have? How about reservoirs? Why is this important?
    - The system has 5 cylinders and 1 reservoir.
    - This is important because it enables each toe break and the parking brake to all work independently. That being said, if the system runs out of hydraulic fluid, none

of the brakes will operate.

- What type of hydraulic fluid does the system use?
  - MIL-H-5606
- Electrical.
  - What are the components of the electrical system?
  - How does the alternator work?
  - How many amps and volts does the alternator produce?
  - How many amps and volts does the battery have?
  - Why does the alternator have 2 more volts than the battery?
  - What does the voltage regulator do? What about the overvoltage relay?
  - How many times can you reset a circuit breaker?
  - Can the engine start if the battery dies?
- Avionics.
- Pitot-static, and associated instruments.
- Vacuum/Pressure, and associated instruments.
- Environmental.
- Deicing and anti-icing.
  - What deicing equipment do we have on our aircraft?
    - Pitot heat
    - Carburetor heat
    - Defrost
  - What anti-icing equipment do we have on our aircraft?
    - Pitot heat
    - Carburetor heat
    - Defrost
  - How does carburetor heat work in our aircraft?
  - How does pitot heat work in our aircraft?
- Oxygen system.
  - What are the oxygen requirements that are applicable to our aircraft?
- Indications of and Managing System Abnormalities or Failures.

Human Factors

- Symptoms, Recognition, Causes, Effects, and Corrective Actions Associated with Aeromedical Issues Including:
  - AIM 8-1
  - Hypoxia.
  - Hyperventilation.
  - Middle ear and sinus problems.
  - Spatial disorientation.
  - Motion sickness.

    - Carbon monoxide poisoning.
    - Stress.
    - Fatigue.
    - Dehydration and nutrition.
    - Hypothermia.
    - Optical Illusions.
        - ICEFLAGS
    - Dissolved nitrogen in the bloodstream after scuba dives.
- Regulations Regarding the Use of Alcohol and Drugs.
- Effects of Alcohol, Drugs, and Over-The-Counter Medications.
- Aeronautical Decision-Making. (ADM)
    - IMSAFE
        - AIM 8-1.
        - **I**llness.
        - **M**edication.
        - **S**tress.
        - **A**lcohol.
        - **F**atigue.
        - **E**motion/ **E**ating.
    - PAVE
        - **P**ilot In Command.
        - **A**ircraft.
        - En**V**ironment.
        - **E**xternal Pressures.
    - 5 Ps
        - **P**lan
        - **P**lane
        - **P**ilot
        - **P**assengers
        - **P**rogramming
    - DECIDE
        - **D**etect
        - **E**stimate
        - **C**hoose
        - **I**dentify
        - **D**o
        - **E**valuate
    - Hazardous Attitudes
        - What are the main hazardous attitudes as identified by the FAA?
            - Anti-Authority.

        - Impulsivity.
        - Invulnerability.
        - Macho.
        - Resignation.
    - What are the antidotes for these hazardous attitudes?
        - The antidote for anti-authority is: “Follow the rules. They are usually right.”
        - The antidote for impulsivity is: “Not so fast. Think first.”
        - The antidote for invulnerability is: “It could happen to me.”
        - The antidote for macho is: “Taking chances is foolish.”
        - The antidote for resignation is: “I’m not helpless. I can make a difference.”

Regulations Applicable to the Private Pilot

- Part 1: definitions and abbreviations.
- Part 23: airworthiness standards for normal category airplanes.
- Part 43: Maintenance, preventive maintenance, rebuilding, and alteration.
- Part 61: Certification: Pilots, flight instructors, and ground instructors.
    - Subpart A: General
    - Subpart B: Ratings
    - Subpart C: Student Pilot
    - Subpart D: Recreational Pilot
    - Subpart E: Private Pilot
    - Subpart F: Commercial Pilot
    - Subpart G: ATP
    - Subpart H: CFI
- Part 67: Medical standards and certification.
- Part 71: Designation of class A, B, C, D, and E airspace areas; air traffic service routes; and reporting points.
- Part 73: Special use airspace.
- Part 91: General operating and flight rules.
- Part 141: Pilot schools.
    - Differentiate between part 61 and part 141.
- Part 61
    - Required documents - 61.3
    - Pilot certificate, medical, and government photo identification when operating as a required pilot.
    - Drugs and alcohol - 61.15 and 61.16
    - Student pilot certificate duration - 61.19
    - Medical certificate classes and durations - 61.23
    - Type rating of aircraft 61.31
    - Pilot Logbooks - 61.51

- Medical Deficiency - 61.53
- Flight Review - 61.56
- Recent flight experience - 61.57
- Change of address - 61.60
- Solo requirements - 61.87
- General limitations (SP) - 61.89
- Solo operations in class b airspace - 61.95

- Part 91
  - Pilot in Command (PIC) responsibilities - 91.3
  - Aircraft airworthiness - 91.7
  - Careless and reckless operation - 91.13
  - Dropping objects - 91.15
- Drugs and alcohol - 91.17
- Preflight action - 91.103:
  - **N**OTAMs.
  - **W**eather.
  - **K**nown ATC delays.
  - **R**unway lengths.
  - **A**lternates if needed.
  - **F**uel requirements.
  - **T**akeoff/landing data.
- Right of way 91.113
- Speed limits 91.117
- Minimum safe altitudes 91.119
- Altimeter Settings 91.121
- Compliance with ATC 91.123
- Temporary Flight Restrictions (TFRs) - 91.137-145
- FAA fuel reserves 91.151
- VFR cruising altitudes 91.159
- Required aircraft equipment 91.205
- Emergency Locator Transmitters 91.207
- Aircraft lights 91.209

Night Preparation

- Definitions of nighttime
  - FAA definition (FAR Part 1)
    - for logging nighttime
    - end of evening civil twilight to the beginning of morning civil twilight.
    - Civil twilight is defined in the Airman's Almanac.

- A much more readable and useful source to find this information is airnav.com on the right-side bar below the airport diagram.
- Position lights (91.209)
  - must be on between sunset and sunrise.
- Nighttime passenger currency (FAR Part 61 - 61.57): currency can be accomplished between 1 hour after sunset and 1 hour before sunrise. The minimum of 3 landings must be to a full stop.

- Night vision
  - How do the eyes work?
    - Rods and cones:
      - Two types of light nerve endings which transmit messages to the brain via the optic nerve.
        - Cones Responsible for color, detail, and far away objects.
          - The cones are located in the center of the retina.
          - The fovea is where almost all of the light-sensing cells are cones.
            - This is the center of vision.
            - Need light in order to function.
        - Rods Responsible for peripheral vision and vision in dim light.
          - The rods are located around the cones (peripherals).
          - Scan using off center viewing at night.
          - When attempting to find traffic do not stare directly at it, look slightly off to the left or right to allow the rods to see the aircraft.
          - The rods can take approximately 30 minutes to fully adapt to the dark.
            - Therefore, it is important to avoid bright lights before and during flight.
              - This is why red flashlights are recommended during flight, they do not disrupt the rod's dark adaptation as much.
    - Due to the increase in the body's need for oxygen at night (increased eye muscle usage, concentration, etc.), the AIM recommends using supplemental oxygen at or above 5,000ft.
      - AIM 8-1-2.
- Night illusions
  - Autokinesis
    - Caused by staring at a single point of light against a dark background creating

        the illusion that the light appears to move.
    - Prevent this by focusing the eyes on objects at varying distances and avoid fixating.
        - Keep the eyes moving and offset/use peripherals.
  - False horizon
    - Caused when the natural horizon is obscured/not readily apparent (night sky and ocean)
    - Generated by confusing bright stars and city lights.
    - Use and trust your instruments to maintain orientation.
  - Featureless terrain / black hole approach
    - An absence of ground features can create the illusion that the aircraft is higher than it actually is.
    - This results in a tendency to fly a lower than normal approach.
    - Use and trust your instruments to maintain orientation.
  - Ground lighting
    - Regularly spaced lights along a road/highway etc. can appear to be runway lights.
    - Bright runway or approach lights can create the illusion the airplane is closer to the runway.
    - Mitigate this as much as possible by maintaining situational awareness.
        - Know what to expect to see (type of airport/runway lighting), where you expect to see it and know where you are (use navaids, GPS, landmarks, etc).
- Lighting
  - Aircraft lights:
    - 91.205 requires the airplane to be equipped with the following lights at night:
        - Landing light (electric) - if the operation is for hire.
        - Anti-collision lights.
        - Position lights.
  - Airport lighting (AIM 2-1):
    - Taxiway lighting:
        - Edge lights - blue.
        - Centerline lights - green, alternates green/yellow on the runway (lead off lights).
        - Runway guard lights - yellow.
        - Taxiway clearance bar lights - yellow.
    - Approach lighting:
        - VASI/PAPI: red and white.
        - Refer to chart supplement to see what approach lighting system and glidepath indicator an airport is equipped with.

        - Runway lighting:
            - Entry threshold - green.
            - Runway edge lights - white.
                - Last 2,000ft - amber.
            - Centerline lights - white.
                - Last 3,000ft - red and white.
                - Last 1,000ft - red.
            - Runway touchdown zone lights - white.
            - Runway end identifier lights - red.
- Preparation and required equipment:
    - Preflight inspection - 91.103:
        - Required equipment for VFR flight at night:
            - ATOMATOFLAMES and FLAPS 91.205 Parts B and C.
        - Pilot equipment:
            - Flashlight: red and white
                - White light is used to preflight the airplane while the red light is used while in flight.
                - Again, red light will not impair night vision.
        - Walk around
            - The preflight inspection is still necessary.
            - White light flashlight is good for the inspection (red light inside the cockpit).
            - Check all aircraft lights.
            - Check the ramp for obstructions.
            - Check NOTAMs, weather, obstructions, navigation systems and radios.
            - Fuel minimums at night are 45 minutes (FAA) and 60 minutes (FIT).
        - In-flight orientation:
            - Relying on instruments at night is needed due to the absence of visual references.
            - Be familiar with runway/airport lights: know what you are looking for!
            - Use the HSI to bug in the runway heading.
            - Don't be afraid to ask ATC for help.
            - Pilot Controlled Lighting:
                - High Intensity Runway Lighting 7 clicks.
                - Medium Intensity Runway Lighting 5 clicks.
                - Low Intensity Runway Lighting 3 clicks.
            - Round out and flare:
                - Judgment of height, speed, and sink rate may be impaired.
                    - There is often a tendency to round out too high.
                - A good rule is to start the round out when the landing lights reflect

on the tire marks on the runway.

Positive Aircraft Control

- Aviate, navigate, communicate.
- Situational Awareness.
- Positive exchange of flight controls.

Land and Hold Short Operations (LAHSO)

- Land and Hold Short Operations (LAHSO).
    - Who can accept it?
    - Do you have to accept it?
    - When can you deviate after accepting?
- Review wire strike avoidance.
    - AIM 7-6-3 dictates that avoidance of guy wires is very difficult due to the visibility of said wires
    - When flying at or below the altitude of antenna towers, the horizontal separation should be no less than 2000 feet.

Controlled Flight into Terrain (CFIT)

- Controlled Flight into Terrain (CFIT).
    - Unintentional collision with terrain (ground, mountain, water, or obstacle).
    - Commonly occurs on approach or during the landing phase of a flight.
    - How does it happen?
        - Loss of situational awareness.
        - Distractions.
        - Procedural mistakes (descent below MDA while IFR).
        - Aircraft performance (high density altitude, tailwind approaches, etc.).
    - How to avoid CFIT?
        - Read AIM 7-6-3 and AIM 7-6-4.
        - Flight Risk Assessment Tool (FRAT).
        - PAVE and IMSAFE acronyms.
        - Using all available resources.
        - Staying vigilant to changing conditions.

Wire Strike Avoidance

- AIM 7-6-3 dictates that avoidance of guy wires is very difficult due to the visibility of said wires
- When flying at or below the altitude of antenna towers, the horizontal separation should be no less than 2000 feet.

Aviation Security

- Aviation Security.
    - Ensure that the aircraft is properly secured after a flight:
        - Tie downs are on and tight.
        - Door and storm window closed.
        - Baggage compartment closed.
    - If you see something, say something!
        - AOPA airport watch: https://www.aopa.org/advocacy/airports-and-airspace/security-and-borders/airport-watch-security
        - Hotline:
            - 866-427-3287.
            - This provides a fast and easy way to report possible threats.

Single-Pilot Resource Management (SRM)

- This starts before the preflight stage.
    - IMSAFE
- 5'Ps
    - Understand the resources available to ensure a safe flight.
- Ensure the rewards outweigh the risks.

Visual Scanning and Collision Avoidance

- It is the PIC responsibility to avoid traffic.
- How to scan for traffic in daytime vs nighttime conditions?
- Clearing procedures.

Systems and Equipment Malfunctions

- Partial or Complete Power loss Including:
    - Engine roughness or overheating.
    - Loss of oil pressure.
    - Fuel starvation.
- Systems and Equipment Malfunctions Including.
    - Electrical malfunction.
    - Vacuum/pressure and associated flight instrument malfunctions.
    - Pitot/static system malfunctions.
    - Electronic flight deck display malfunction.
    - Landing gear or flap malfunction.
    - Inoperative trim.
- Smoke/Fire/Engine Compartment Fire.

- Any Other System Specific to the Airplane.
- Inadvertent Door or Window Opening.

Emergency Equipment and Survival Gear

- Emergency Locator Transmitter (ELT) Operations, Limitations, and Testing Requirements.
- Fire Extinguisher Operations and Limitations.
- Emergency Equipment and Survival Gear Needed for:
  - Climate extremes. (hot/cold)
  - Mountainous terrain.
  - Overwater operations.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 147.

Completion Standards

Through oral questioning the instructor shall determine that all knowledge areas meet the FAA Private Pilot ACS.